ISE549 – Management of Innovation & Technology

Week 10: Critical Skills for Managing the Innovation Process

Complete English Transcript of the Lecture
(Cleaned & structured from Audio Part 1 / Part 2 + Lecture Slides)

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Sources: ISE549 W10_GN.pdf + ISE549_W10_Part1.m4a + Part2.m4a. Minor transcription corrections for clarity; content remains faithful to the original lecture.

1. Recap: The 3I Path and Continuous Innovation Process

Innovation follows the familiar 3I sequence: **Idea** (ideation and screening) → **Invention** (experimentation, trial-and-error, solution discovery and refinement) → **Innovation** (commercialisation – mass production, distribution, sales & marketing – so that the invention reaches the market).

A more detailed continuous process starts with problem / need identification, moves through idea generation and evaluation, invention and solution selection, implementation and scale-up, and finally impact and commercialisation. Only after successful commercialisation does the invention become a true market innovation.

2. Ansoff Matrix (Product–Market Growth Matrix)

The Ansoff Matrix is a classic marketing and strategy tool that maps growth options on two dimensions: Product (existing ↔ new) and Market (existing ↔ new). It produces four strategies with rising levels of risk and complexity:

- **Market Penetration** (existing product, existing market) – grow share in the current market with current offerings. Lowest risk.
- **Product Development** (new product, existing market) – launch new products to the same customers / market. Leverages existing relationships and knowledge.
- **Market Development** (existing product, new market) – take current products into new geographic markets, new segments or new industries. Requires market knowledge and network building.
- **Diversification** (new product, new market) – both product and market are new. Highest risk and complexity because the firm has neither product nor market familiarity.

Most firms begin in the top-left cell (penetration) and move outward only when they have sufficient capability and appetite for risk. Diversification is usually the last and most demanding step.

2.1 Illustrative Examples

Xiaomi: smartphones in China (penetration) → home appliances and IoT devices to the same customers (product development) → overseas expansion with existing product lines (market development) → electric vehicles (diversification into a new product category and new competitive arena).

Coca-Cola: classic Coke (penetration) → Diet Coke / Coke Zero (product development for different segments) → geographic expansion and new beverage categories → further moves into adjacent or unrelated areas (e.g., fashion/apparel experiments).

McDonald's: core menu and QSV (Quality, Service, Value) model (penetration) → global restaurant expansion (market development) → local menu adaptation (e.g., vegetarian options in India) → diversification into hotels and internal/external training universities.

Apple: computers and later iPhone in core markets (penetration) → accessories and additional product lines (product development) → geographic expansion (market development) → highest-risk moves that combine new products with new markets.

3. Organisational Factors that Shape Innovation

Whether a firm produces incremental or radical innovation depends heavily on internal conditions:

- **Organisational culture** – open, aggressive, idea-welcoming cultures versus conservative, bureaucratic, hierarchical cultures.
- **Leadership commitment** – leaders who invest resources, tolerate risk and visibly champion innovation versus leaders who prioritise short-term efficiency.
- **Structure** – tall hierarchical structures tend to slow ideas; flatter, more horizontal structures facilitate faster communication and decision-making.
- **Reward and recognition systems** – formal pathways such as intrapreneurship / corporate entrepreneurship that allow employees to champion ideas, receive sponsorship and potentially lead new ventures. A climate of "no fear of failure" is essential; otherwise potential innovators will not risk their careers.

4. Innovator's Dilemma, Sustaining vs Disruptive Innovation, Red vs Blue Ocean

Successful incumbents can still fail when they focus exclusively on serving existing high-margin customers with sustaining innovations (improvements that keep current competitive advantages). Disruptive innovations often start in lower-end or new segments, rewrite the rules of the game, and eventually overtake the mainstream market.

Red Ocean = existing product/market space that is already crowded and competitive; competition becomes a zero-sum, price-cutting "bloody" fight that erodes margins and reduces willingness to invest in innovation.

Blue Ocean = creation of new uncontested market space through value innovation; the firm makes competition irrelevant by offering a new combination of value. Start-ups frequently pursue blue-ocean strategies because they have little existing infrastructure to protect and therefore face a "do-or-die" incentive to create something new.

Incumbents need sustaining innovation to defend current advantages, while also finding ways (internally or through separate units) to explore disruptive possibilities.

5. Global Strategic Orientations (linked to standardisation vs local adaptation)

Firms competing internationally choose different balances between global integration and local responsiveness:

- High global standardisation / low local adaptation (e.g., many pharmaceutical products, heavy equipment such as Caterpillar) → global strategy.
- High local adaptation / lower global standardisation (e.g., certain consumer appliances) → multi-domestic strategy.
- High on both dimensions → transnational strategy (e.g., 3M).
- Lower on both → international strategy (e.g., classic McDonald's approach in earlier stages).

6. Dominant Design and Winner-Takes-All Markets

A dominant design (or dominant standard) emerges when one technical solution becomes the de-facto industry standard. Once established, network externalities reinforce its lead: more users attract more complementary products and developers, which in turn attract still more users.

Classic examples discussed:

- Colour-television broadcasting standards: NTSC (mainly North America and parts of Asia/Latin America), PAL (much of Europe and Asia), SECAM (France, parts of Africa and the former Soviet sphere). Governments sometimes intervene to choose or protect a standard for compatibility and welfare reasons.
- Personal-computer operating systems: the rise of MS-DOS and later Microsoft Windows created a powerful "winner-takes-most" position. Software developers preferentially wrote for the largest installed base, further reinforcing Windows' dominance.
- Digital audio: progression from physical media through MP3 players to smartphone-based streaming platforms (Spotify, Apple Music, etc.).

Benefits of a large network (compatibility, rich complements, lower effective cost for users) trade off against the costs of monopoly power (reduced consumer choice, potential for higher prices, slower innovation by the dominant firm). Microsoft, for example, has historically priced Windows below the theoretical monopoly maximum in order to grow and protect the ecosystem.

7. Government Responses to Market Dominance / Monopoly

When dominance becomes excessive, governments may intervene through:

- Antitrust / competition legislation and large fines.
- Direct price controls (e.g., fuel pricing).
- Support for consumer associations and public scrutiny.
- Creation or enforcement of fair-competition regimes (e.g., telecommunications regulation).
- In extreme cases, nationalisation or taking assets back into public ownership at the end of franchise periods (illustrated by Hong Kong cross-harbour tunnels).

8. Protecting Intellectual Property (IP)

Three primary legal instruments (plus trade secrets) are used to protect the fruits of invention and creativity:

Patents – grant the owner the exclusive right to make, use or sell an invention for a limited period (commonly 20 years). They cover processes, machines, manufactured articles, compositions of matter, ornamental designs and certain plants. The inventor must fully disclose how the invention works. Licensing allows others to use the patented technology in exchange for royalties. Example discussed: Pfizer's Viagra – originally developed for other indications, the unexpected effect on erectile dysfunction led to a highly valuable patented product whose exclusivity later expired, opening the market to generics.

Trademarks – protect distinctive signs (words, logos, slogans, colours, sounds, trade dress / look-and-feel) that identify the source of goods or services. Famous examples: Nike "swoosh" and "Just Do It", Apple logo. Trademark rights are territorial; the Ferrero Rocher / "Jinsha" case in China illustrates how a local registrant can block a foreign brand's preferred Chinese name inside that jurisdiction.

Copyright – protects original literary, artistic, musical and certain other works from unauthorised reproduction, adaptation, public performance or display. Fair-use / fair-dealing exceptions exist for criticism, news reporting, teaching and research. Students are reminded to cite sources properly when using copyrighted material in assignments.

Trade secrets – commercially valuable information that is kept confidential and derives value from not being generally known. No registration is required, but reasonable secrecy measures must be maintained.

Historical illustration: the "War of the Currents" between Thomas Edison (champion of direct current, DC) and Nikola Tesla / George Westinghouse (champions of alternating current, AC). AC ultimately became the dominant standard for power transmission, yet both AC and DC continue to coexist in modern systems. Edison held an extraordinary number of patents across many fields, demonstrating the long-term value of systematic IP protection.

9. Exam-Friendly Key Takeaways

- Innovation process = Idea → Invention → Innovation (commercialisation). Always start from a real problem or market need.
- Ansoff Matrix: Penetration → Product Development → Market Development → Diversification (rising risk).
- Internal enablers of innovation: culture, leadership commitment, flat structure, reward systems that tolerate failure (intrapreneurship).
- Sustaining innovation defends current advantages; disruptive innovation rewrites the rules and can dethrone incumbents. Red Ocean = bloody existing competition; Blue Ocean = new uncontested space.
- Dominant designs create winner-takes-most dynamics via network externalities; governments may intervene when monopoly power becomes excessive.
- Core IP tools: Patents (exclusive right to invention), Trademarks (source identity), Copyright (creative works), Trade secrets (confidential know-how). Protection is territorial and time-limited; licensing monetises the rights.

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